

Lecture 22: Image diffusion and multimodal generation

PSYC 51.17: Models of language and communication

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Learning objectives

By the end of this lecture, you will be able to...

1. Explain how **latent diffusion** compresses computation via a VAE to enable high-resolution image generation
2. Describe how **CLIP** creates a shared embedding space for text and images
3. Explain how **cross-attention** and **classifier-free guidance** enable text-controlled image generation
4. Compare the **U-Net** and **Transformer (DiT)** backbones for diffusion models
5. Generate images from text prompts using the HuggingFace `diffusers` library

From text to images

Bridging Lecture 21 and today

In Lecture 21, we saw how diffusion models generate **text** by iteratively unmasking tokens (discrete diffusion). Today we turn to the domain where diffusion was born — **images** — and ask: how do we generate high-resolution images from text prompts?

The key challenges

Challenge	Solution (this lecture)
Images are huge (786K+ values)	Latent diffusion — compress first, diffuse in latent space
Text must control generation	CLIP + cross-attention — bind words to spatial regions
Output must match the prompt	Classifier-free guidance — amplify text alignment
Architecture must scale	U-Net or DiT — backbones for the denoising network

The pixel problem

Why pixel-space diffusion is expensive

The DDPM framework (Lecture 21) can be applied to images by adding Gaussian noise to pixels and learning to reverse the process. But for a 512×512 RGB image, the denoising network must process $512 \times 512 \times 3 = 786,432$ values at every step — far too expensive for high-resolution generation.

U-Net (brief preview)

The **U-Net** ([Ronneberger et al., 2015](#)) is an encoder-decoder neural network with skip connections, originally designed for image segmentation. It became the default backbone for diffusion models — we'll cover it in detail later this lecture.

The numbers

Resolution	Pixels	Memory	Time per image
64×64	12,288	~2 GB	~30 seconds
256×256	196,608	~8 GB	~5 minutes
512×512	786,432	~32 GB	~20 minutes

The solution: don't run diffusion in pixel space. Run it in a **compressed latent space**.

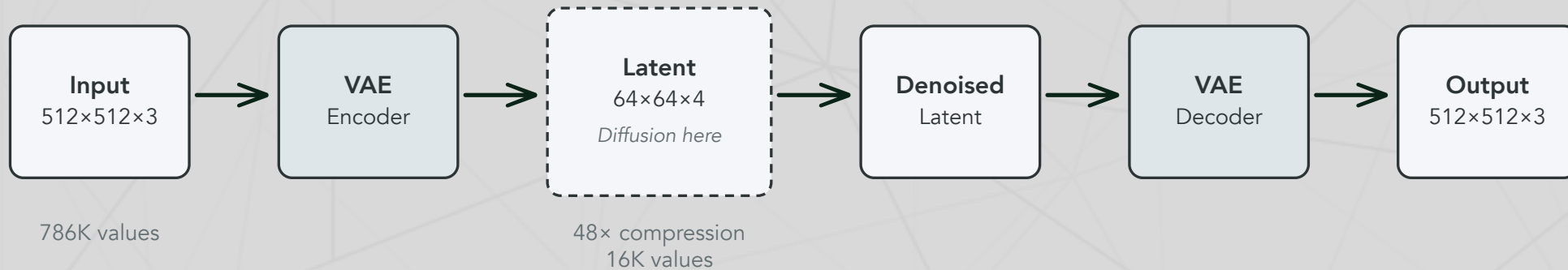
Latent diffusion

Compressing before diffusing

Latent diffusion separates image generation into two stages:

1. **Compression:** A pretrained **variational autoencoder** (VAE) encodes images into a compact latent space (typically $8\times$ spatial compression)
2. **Generation:** The diffusion process operates entirely in the latent space before decoding back to pixels

The VAE is described on the next slide.



Further reading

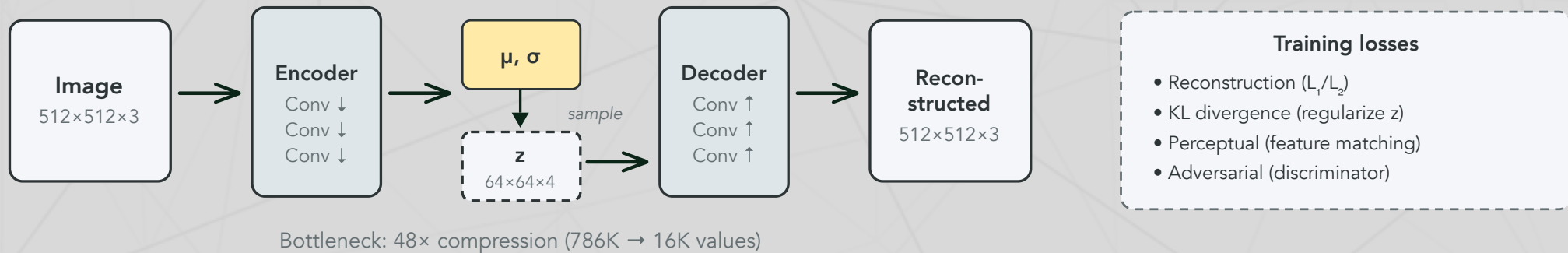
Rombach et al. (2022, CVPR). "High-resolution image synthesis with latent diffusion models" — The paper that introduced latent diffusion and enabled Stable Diffusion.

Variational autoencoder (VAE)

Compressing images to a latent space

A **VAE** learns to compress images into a low-dimensional latent representation and reconstruct them:

1. **Encoder:** Maps image $\mathbf{x} \in \mathbb{R}^{512 \times 512 \times 3}$ to mean $\boldsymbol{\mu}$ and standard deviation $\boldsymbol{\sigma}$ of a latent distribution
 - $\boldsymbol{\mu}$ represents the "average" latent representation for the image
 - $\boldsymbol{\sigma}$ captures uncertainty — how much the latent can vary while still reconstructing the image
2. **Sampling:** Draw $\mathbf{z} \sim \mathcal{N}(\boldsymbol{\mu}, \boldsymbol{\sigma}^2)$ — a latent vector in $\mathbb{R}^{64 \times 64 \times 4}$
3. **Decoder:** Reconstructs the image from $\mathbf{z}$



Why this works

The VAE learns to preserve perceptually important information while discarding redundant detail. Diffusion in latent space is **~50× cheaper** than in pixel space, with negligible quality loss. This single insight enabled Stable Diffusion — the first open-source, consumer-GPU image generator.

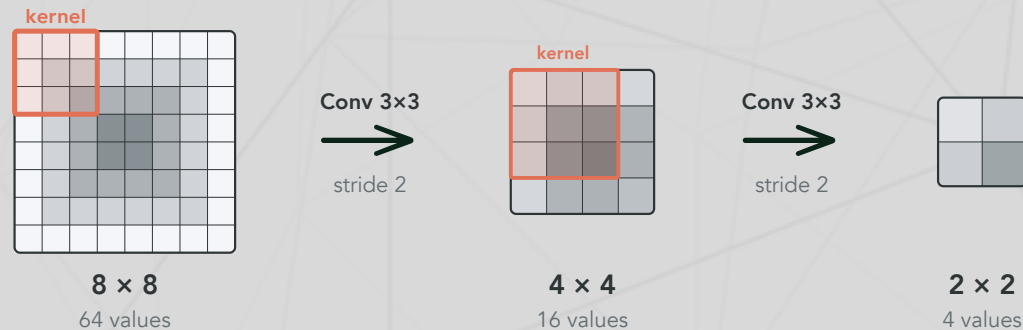
How convolutions compress images

Convolution: the key building block

A **convolution** slides a small filter (kernel) across an image, computing a weighted sum at each position. With **stride 2**, the filter moves 2 pixels at a time, halving the spatial dimensions:

- **8×8** input → 3×3 conv, stride 2 → **4×4** output (4× fewer values)
- **4×4** → 3×3 conv, stride 2 → **2×2** (4× fewer again)

Stacking convolution layers creates a hierarchy: pixels → edges → textures → objects.



3×3 averaging kernel (weight = 1/9)

Stride 2: kernel moves 2 cells per step, halving dimensions

8×8 → 4×4: each output cell = average of a 3×3 patch

4×4 → 2×2: second pass further compresses spatial info

Cell shading shows computed convolution values — darker cells correspond to higher-intensity input regions.

The intuition

In practice, kernels are **learned**, not fixed. Early layers detect **edges** and gradients; deeper layers compose these into **textures** and **shapes**. The averaging kernel above is simplified — real networks learn *what* to compress. This hierarchy (pixels → edges → textures → objects) is exactly what the VAE encoder learns.

The VAE bottleneck

Stable Diffusion's VAE in numbers

Property	Value
Input resolution	$512 \times 512 \times 3$
Latent resolution	$64 \times 64 \times 4$
Compression ratio	$48\times$ (786K $\rightarrow$ 16K values)
Reconstruction quality	Near-lossless for natural images

The VAE is trained once and frozen. The diffusion model only ever sees latent representations — it never touches pixels during training or sampling.

How the VAE is trained

- **Reconstruction loss:** Ensure decoded image matches the original
- **KL divergence:** Regularize the latent space so it's smooth and continuous
- **Perceptual loss:** Compare *features* extracted by a pretrained network (e.g., VGG), not raw pixels — so the model prioritizes visual similarity over pixel-exact matching
- **Adversarial loss:** A discriminator network tries to distinguish real from reconstructed images, pushing the decoder toward photorealistic outputs

KL divergence

The **Kullback-Leibler (KL) divergence** measures how one probability distribution Q diverges from a reference distribution P :

$$D_{KL}(Q\|P) = \int Q(z) \log \frac{Q(z)}{P(z)} dz$$

Intuitively, it quantifies how much information is lost when Q is used to approximate P . In VAEs, we want the learned latent distribution to be close to a simple prior (e.g., standard normal) to ensure smoothness and generalization.

CLIP: connecting text and images

Contrastive language-image pre-training (CLIP)

CLIP learns a **shared embedding space** for text and images. It was trained on 400 million image-text pairs from the internet:

1. **Image encoder** (Vision Transformer): Maps an image to a vector
2. **Text encoder** (Transformer): Maps a caption to a vector
3. **Contrastive training**: Matching pairs are pulled close; non-matching pairs are pushed apart

Why CLIP matters for diffusion

CLIP provides the "language understanding" for text-to-image systems. When you type "a cat wearing a hat," CLIP's text encoder converts it into an embedding that captures the **visual meaning** — not just the linguistic meaning. This embedding then guides diffusion via cross-attention.

Further reading

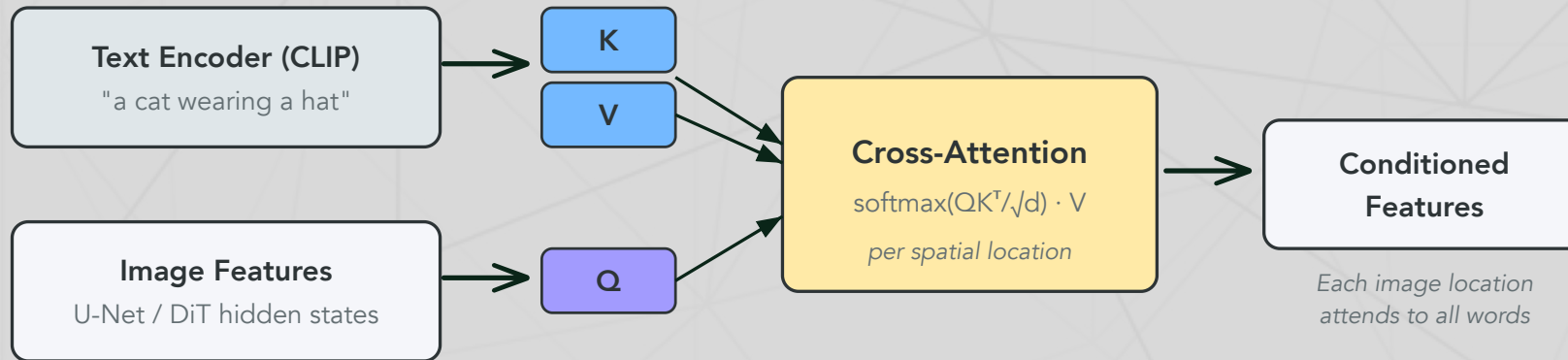
Radford et al. (2021, ICML) "Learning transferable visual models from natural language supervision" — Connection to Lecture 11: CLIP extends the idea of **learned embeddings** (Word2Vec, GloVe) to a *joint* text-image space.

Text conditioning with cross-attention

How text controls image generation

To generate images from text, the denoising backbone adds **cross-attention** layers (recall attention from Lecture 15). At each layer:

1. The text prompt is encoded by CLIP into a sequence of token embeddings
2. The image features at that layer serve as **queries** (Q)
3. The text embeddings serve as **keys** (K) and **values** (V)
4. Cross-attention allows each spatial location in the image to attend to relevant words



How 'a cat wearing a hat' becomes an image

The word "cat" activates high attention weights in the spatial region where the cat is generated. The word "hat" activates weights near the top of the cat region. This spatial-linguistic binding is learned entirely from image-caption pairs.

Classifier-free guidance

Controlling generation quality

Classifier-free guidance (CFG) improves alignment between text and generated images. During training, the text condition is randomly dropped some fraction of the time. At inference:

$$\tilde{\epsilon} = \underbrace{\epsilon_{\theta}(\mathbf{x}_t, t, \emptyset)}_{\text{unconditional}} + w \cdot \left(\underbrace{\epsilon_{\theta}(\mathbf{x}_t, t, c)}_{\text{conditional}} - \underbrace{\epsilon_{\theta}(\mathbf{x}_t, t, \emptyset)}_{\text{unconditional}} \right)$$

- ϵ_{θ} — the **denoising network** (U-Net or DiT), parameterized by θ
- $\mathbf{x}_t$ — the **noisy latent** at timestep t
- c — the **text condition** (CLIP embedding of the prompt)
- $\emptyset$ — **no text** (empty prompt, i.e., unconditional)
- **Unconditional** $\epsilon_{\theta}(\mathbf{x}_t, t, \emptyset)$ — what the model predicts *without* any text guidance
- **Conditional** $\epsilon_{\theta}(\mathbf{x}_t, t, c)$ — what it predicts *guided by* the prompt
- w — the **guidance scale** (typically 7–15): how much to amplify the text signal

The intuition

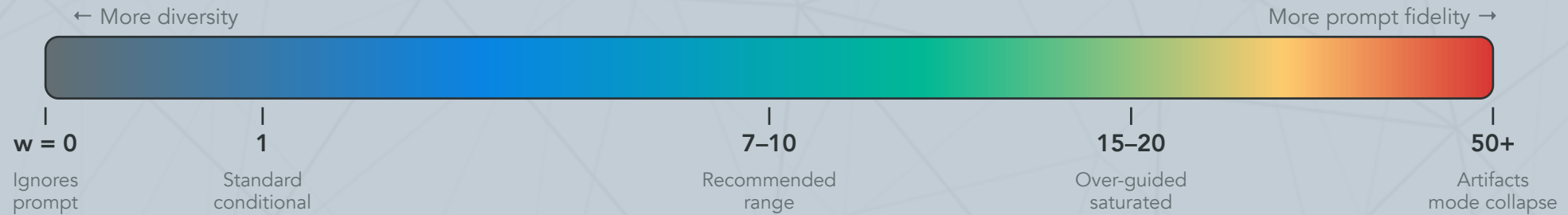
Think of CFG as asking: "What's *different* about images that match this prompt versus random images?" Then amplifying that difference. The model learns both what a "dog on a beach" looks like *and* what a random image looks like — guidance amplifies the gap. See the next slide for examples.

Further reading

Ho & Salimans (2022), "Classifier-free diffusion guidance" — Used in virtually all modern text-to-image systems.

The guidance scale tradeoff

What different guidance scales produce



In practice

Nearly every text-to-image system uses CFG. Stable Diffusion defaults to $w = 7.5$; DALL-E 2 uses $w \approx 4$. The range of 7-10 balances prompt fidelity against visual quality — pushing higher creates oversaturated, artifact-prone images.

Try it: varying the guidance scale

Experiment with CFG (Google Colab)

```
1 from diffusers import StableDiffusionPipeline
2 import torch
3
4 pipe = StableDiffusionPipeline.from_pretrained(
5     "runwayml/stable-diffusion-v1-5", torch_dtype=torch.float16
6 ).to("cuda")
7
8 for scale in [1.0, 7.5, 20.0]:
9     img = pipe("a cat in a hat", guidance_scale=scale).images[0]
10    img.save(f"guidance_{scale}.png")
```

What to expect

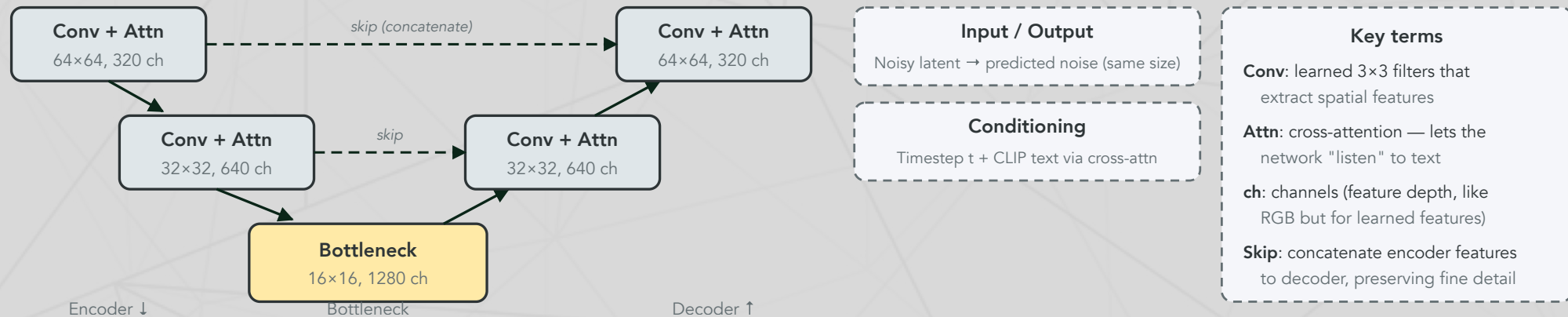
- **scale = 1.0:** Effectively no guidance — diverse but often off-prompt
- **scale = 7.5:** Default sweet spot — faithful to prompt with natural variation
- **scale = 20.0:** Over-guided — saturated colors, sharp artifacts, but very literal

U-Net: the original diffusion backbone

An encoder-decoder with skip connections

The U-Net is the default backbone for diffusion models from DDPM through Stable Diffusion 1–2. Its U-shaped design provides **multi-scale reasoning**:

1. **Encoder** (downsampling): Reduces spatial resolution while increasing channels — captures global context
2. **Bottleneck**: Lowest resolution, largest receptive field — reasons about overall composition
3. **Decoder** (upsampling): Restores spatial resolution — generates fine details
4. **Skip connections**: Concatenate encoder features directly to the matching decoder layer — without these, the decoder must reconstruct spatial detail from the bottleneck alone, losing fine texture and edges



Further reading

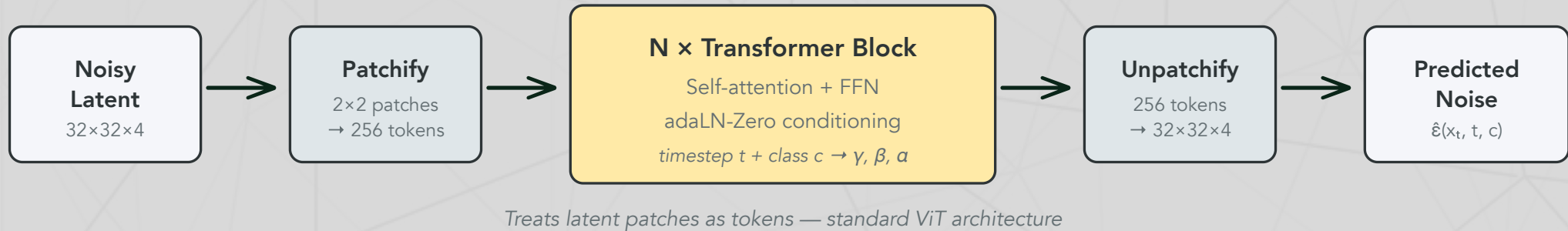
Ronneberger et al. (2015, *MICCAI*) "U-Net: convolutional networks for biomedical image segmentation" — Originally for medical imaging, now the workhorse of diffusion.

Diffusion Transformer (DiT)

Replacing U-Net with a Transformer

The Diffusion Transformer (DiT) replaces the U-Net with a standard Vision Transformer:

1. **Patchify**: Divide the noisy latent into non-overlapping patches (e.g., 2×2)
2. **Flatten**: Treat patches as a sequence of tokens (just like ViT treats image patches — Lecture 15)
3. **Process**: Apply standard Transformer blocks with self-attention
4. **Unpatchify**: Reshape back to spatial dimensions to get the predicted noise



Key terms

- **Noisy latent**: The VAE-compressed image with Gaussian noise added at timestep t — this is the input the denoising network must "clean up"
- **Predicted noise**: The network's estimate of what noise was added — subtract it from the noisy latent to get a cleaner image

DiT: why replace U-Net?

Transformers scale better

DiT-XL/2 (675M parameters) achieves state-of-the-art FID of 2.27 on ImageNet, beating all previous diffusion models. More importantly, DiT shows **clean scaling behavior** — larger models consistently produce better results, with no architectural bottlenecks.

Key terms

- **Fréchet Inception Distance (FID)** (Heusel et al., 2017) measures the distance between real and generated image distributions using features from a pretrained Inception network. Lower FID = more realistic.
- **Inductive biases** are assumptions built into the architecture — e.g., convolutions assume local spatial structure; Transformers make fewer such assumptions and let the model learn structure from data.

The intuition

U-Nets have strong inductive biases for spatial data (locality, hierarchy). These help with small models but become constraints at scale. Transformers make fewer assumptions — the same lesson we saw with language models (Lectures 15–16).

Further reading

Peebles & Xie (2023, ICCV) "Scalable diffusion models with Transformers" — DiT showed that Transformers can replace U-Nets in diffusion, and the result scales better.

Try it: generate an image

Text-to-image with Stable Diffusion (Google Colab)

```
1 !pip install -q diffusers transformers accelerate
2
3 import torch
4 from diffusers import StableDiffusionPipeline
5
6 pipe = StableDiffusionPipeline.from_pretrained(
7     "runwayml/stable-diffusion-v1-5", torch_dtype=torch.float16
8 ).to("cuda")
9
10 image = pipe("a photo of an astronaut riding a horse on mars").images[0]
11 image.save("astronaut.png")
```

What's happening under the hood

1. CLIP encodes your text prompt into embeddings
2. The U-Net iteratively denoises a random latent, guided by those embeddings via cross-attention
3. CFG amplifies the text signal at each step (default `guidance_scale=7.5`)
4. The VAE decoder converts the final latent back to a 512×512 pixel image

Every concept from this lecture is working together in those 5 lines of code.

Take-home messages

Key ideas from today

- The key bottleneck in high-resolution generation wasn't the diffusion process itself — it was **where** you run it. Compressing to latent space via a VAE made consumer-GPU generation possible.
- Text-to-image generation requires a **bridge between modalities**: CLIP creates a shared embedding space (extending the idea of word embeddings from Lecture 11), and cross-attention lets the diffusion model "listen" to text at every spatial location (using the same mechanism as encoder-decoder attention from Lecture 15).
- Classifier-free guidance shows that **controlling** generation is as important as generation itself — the trick is surprisingly simple: learn what conditional and unconditional outputs look like, then amplify the difference.
- The field evolves by **composing** innovations (VAE + CLIP + CFG + U-Net/DiT), not replacing them. Each addresses one specific limitation.

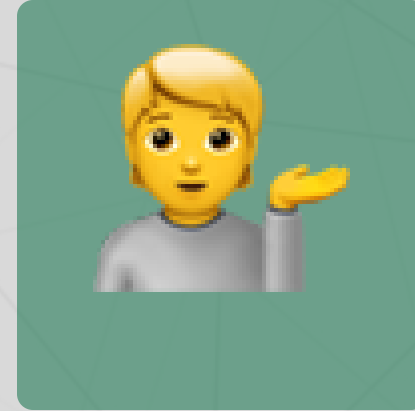
Questions?



Email



Discord



Office hours

Up next...

Diffusion applications: text-to-video (Sora), text-to-audio, discrete diffusion for text, and the ethics of generative AI